Beyond Rollups: Evaluating EIP-4844 Blobs as Temporary Verifiable Data Availability for Research Artifacts

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Abstract

Data availability is foundational to reproducible computational science, yet existing decentralized storage primitives force researchers to choose between expensive, synchronous on-chain persistence and unverifiable off-chain pinning. Ethereum EIP-4844 introduced transient data blobs as an economical consensus mechanism, but its design was targeted almost exclusively at Layer-2 transaction rollups. In this paper, we explore a non-rollup application model: using EIP-4844 blobs as a temporary, verifiable publication layer for structured scientific and machine-learning research artifacts. We design and implement an end-to-end reproducible pipeline featuring RFC 8785 canonical record serialization, modulus-safe field element packing (`FE31`), a three-tiered commitment architecture combining blob versioned hashes, whole-dataset content hashes, and record-level Merkle proofs, and a dual-horizon content-addressed archival bridge. We empirically evaluate our system across five real-world and synthetic dataset classes (OpenAlex scholarly metadata, NOAA climate observations, PyTorch Transformer training logs, synthetic scaling traces, and incompressible controls) up to 50 MiB. Our results demonstrate a 27.8× theoretical uncompressed cost reduction and up to 43.8× measured compressed cost reduction over standard EVM calldata, with encoding throughput exceeding 170 MiB/s on commodity workstations. However, an empirical multi-tier retrieval study on Ethereum Sepolia reveals that public consensus Beacon endpoints exhibit severe instability and rate-limiting for historical sidecars. We demonstrate that while Ethereum blobs provide an economical, consensus-committed publication window, durable long-term scientific preservation strictly requires a decoupled, content-addressed archival bridge. Our full artifact, contracts, and evaluation harnesses are open source and independently reproducible.

1 Introduction

The reproducibility of empirical scientific research depends directly upon the integrity and verifiable availability of experimental datasets, hyperparameter configurations, and model training traces [11]. In computational disciplines such as machine learning and environmental informatics, research artifacts frequently consist of multi-megabyte structured logs. When these artifacts are hosted on traditional institutional web servers or cloud storage buckets, they remain vulnerable to link rot, silent bit rot, domain lapses, and unilateral tampering.

Decentralized networks have emerged as a potential solution, but existing blockchain primitives present an acute economic and architectural dilemma:

1. **EVM State Storage (`SSTORE`)**: Storing raw dataset records directly within persistent smart contract storage slots costs upwards of $2,900 USD per megabyte under typical gas regimes, rendering large-scale scientific publication economically impossible.
2. **Standard Transaction Calldata**: Calldata commits historical byte streams into full node block history, but incurs substantial execution gas costs ($1.44 USD/MiB), which will further increase under proposed calldata pricing floors such as EIP-7623 [10].
3. **Off-Chain P2P Storage (IPFS, Filecoin, Arweave)**: While content-addressed networks provide cheap storage, raw IPFS hashes lack native consensus availability verification on Ethereum L1, and storage deals require separate tokenomics or ongoing subscription pinning [2, 7].

In March 2024, the Ethereum Deneb hard fork activated EIP-4844 (Proto-Danksharding) [3]. EIP-4844 introduced "blobs"—128 KiB binary data sidecars that are verified by consensus validators via Kate-Zaverucha-Goldberg (KZG) polynomial commitments [5] and made available for a temporary window of 4,096 epochs (≈ 18.2 days) before automatic pruning. While EIP-4844 was engineered specifically to reduce Layer-2 (L2) rollup settlement fees, its core properties—consensus-enforced data availability, low gas fees, and cryptographic commitment anchoring—suggest an unexplored application: *temporary, verifiable publication for structured scientific research artifacts.*

1.1 Central Research Question and Contributions

This paper addresses the central question:

> *Can Ethereum blobs serve as a temporary, verifiable publication layer for reproducible research artifacts, and what additional infrastructure is required to convert temporary data availability into durable scientific preservation?*

We make the following contributions:

1. **Non-Rollup Application Model**: We formulate and implement an end-to-end architecture applying EIP-4844 blobs to scientific data publication, record streaming, and peer review.

2. **Deterministic Encoding Pipeline**: We design a deterministic pipeline incorporating RFC 8785 JSON Canonicalization Scheme (JCS) [9], Zstandard compression (RFC 8878) [4], and modulus-safe field element modulus packing (`FE31`).

3. **Three-Tiered Hybrid Commitments**: We implement an EVM contract suite (`DatasetRegistry.sol`) that binds blob versioned hashes (`BLOBHASH`), whole-dataset canonical content hashes, and position-binding record-level Merkle proofs for sub-record verification and optimistic dispute games.

4. **Dual-Horizon Archival Bridge**: We design, deploy, and verify an archival extension contract (`ResearchArchiveRegistry.sol`) and client bridge (`archival_bridge.py`) that binds transient consensus blob commitments to durable content-addressed storage (IPFS/CAS), ensuring bitwise verification post consensus pruning.

5. **Empirical Infrastructure Realities**: We conduct an empirical retrieval study on Ethereum Sepolia, documenting practical limitations, rate-limiting, and failure modes across public consensus Beacon API endpoints.

6. **Guarantee-Aware Comparative Framework**: We formalize a multi-dimensional comparison evaluating availability duration, consensus verification, permanence, retrieval guarantees, and economics across seven decentralized and cloud storage baselines.

7. **Reproducible Open-Source Artifact**: We release verified Sepolia contracts, automated reproduction harnesses, raw trial-level benchmarks, and a formal security threat model.

2 Background and Protocol Fundamentals

2.1 EIP-4844 and Proto-Danksharding

Ethereum EIP-4844 introduced Type-3 transactions carrying opaque binary payloads known as *blobs* [3]. Under the initial Cancun/Deneb activation, blocks target 3 blobs (384 KiB) with a hard maximum of 6 blobs (768 KiB). Unlike standard transaction calldata, blob payloads are not accessible directly to the Ethereum Virtual Machine (EVM) via execution opcodes, nor are they retained in perpetual execution state. Instead, execution contracts interact with blobs exclusively through the `BLOBHASH` opcode, which returns the 32-byte versioned hash of a blob carried by the current transaction.

2.2 KZG Polynomial Commitments and Field Modulus Constraints

Each blob consists of exactly 4,096 field elements in the scalar field of the BLS12-381 elliptic curve construction. Each field element occupies 32 bytes ($4,096 \times 32 = 131,072$ bytes / 128 KiB). The BLS12-381 scalar field modulus p is defined as:

$$p = \texttt{0x73eda753299d7d483339d80809a1d805}\\ \texttt{53bda402fffe5bfeffffffff00000001} \approx 5.24 \times 10^{76} \tag{1}$$

Because $2^{254} < p < 2^{255}$, arbitrary 32-byte binary sequences may exceed p. If an element $\geq p$ is passed to consensus clients, KZG commitment generation or point evaluation precompile calls revert immediately. Consequently, application-level payloads require modulus-safe packing.

A blob's cryptographic identity is anchored by its KZG commitment $C \in \mathbb{G}_1$. The Ethereum consensus layer derives the versioned hash H_v by taking the SHA-256 digest of C, replacing the first byte with the protocol version identifier `0x01`:

$$H_v = \texttt{0x01} \parallel \text{SHA-256}(C)[1:32] \tag{2}$$

2.3 The 4,096-Epoch Availability Window

Consensus layer clients (Prysm, Lighthouse, Teku, Lodestar, Nimbus) propagate blob sidecars across a dedicated P2P gossipsub network. Crucially, consensus specifications require clients to retain blob sidecars for only 4,096 epochs:

$$\begin{aligned} T_{\text{avail}} &= 4,096 \text{ epochs} \times 32 \text{ slots/epoch} \times 12 \text{ s/slot} \\ &= 1,572,864 \text{ s} \approx 18.2 \text{ days} \end{aligned} \tag{3}$$

After T_{avail}, consensus nodes prune blob sidecars from disk storage. While the on-chain versioned hash remains

permanently stored in historical block headers, the underlying data bytes become unavailable from standard consensus clients.

2.4 Multidimensional EIP-1559 and Blob Gas Economics

EIP-4844 established a separate, independent EIP-1559 fee market for blob gas [8]. A single blob consumes exactly 131,072 blob gas. The blob base fee f_{blob} floats dynamically according to exponential target tracking:

$$f_{\text{blob},n+1} = f_{\text{blob},n} \cdot \exp\left(\frac{\text{BlobGasUsed}_n - \text{TargetGas}}{434,496}\right) \tag{4}$$

When blob demand remains below target capacity (e.g. 3 blobs per block), the blob base fee drops to its minimum threshold of 1 wei (10^{-9} gwei). Under normal non-congested conditions, the total cost of publishing a 128 KiB blob is dominated by the standard execution priority fee rather than the blob gas itself.

3 Related Work

3.1 Modular Data Availability Layers

Modular blockchain architectures decouple transaction execution from consensus and data availability. Celestia [1] implements a sovereign data availability blockchain utilizing two-dimensional Reed-Solomon erasure coding and Namespaced Merkle Trees (NMTs) to enable light clients to perform Data Availability Sampling (DAS). Avail [13] similarly implements a standalone DA substrate using Kate polynomial commitments and erasure coding. EigenDA [8] provides a restaked data availability service on Ethereum, dispersing data slices across an off-chain node quorum. While these systems provide scalable bandwidth, they operate as separate sovereign networks or restaked middleware, requiring external bridges and distinct trust assumptions. In contrast, our approach utilizes native Ethereum L1 blobs, achieving direct consensus commitment without secondary network tokenomics.

3.2 Decentralized Archival Storage

Decentralized storage networks have long sought to preserve digital artifacts:

- **IPFS** [2]: Organizes data into directed acyclic graphs (IPLD) identified by cryptographic content identifiers (CIDs). However, IPFS does not provide consensus-backed persistence guarantees; unpinned data is rapidly discarded by gateway caches.
- **Filecoin** [7]: Introduces an economic market for contractual storage deals verified by Proof-of-Replication (PoRep) and Proof-of-Spacetime (PoSt). While highly durable, Filecoin deals require sector sealing latencies, separate FIL currency management, and periodic deal renewals.
- **Arweave** [12]: Uses a Proof-of-Access consensus mechanism funded by an upfront endowment pool intended to sustain multi-decade data persistence.

In our architecture, these networks are not viewed as competitors to blobs, but rather as complementary backends within a dual-horizon framework: blobs serve as the immediate, consensus-verified publication layer, while IPFS and Arweave provide durable post-expiry preservation.

3.3 Scientific Artifact Provenance and DeSci

The Decentralized Science (DeSci) movement seeks to make scientific workflows transparent, immutable, and verifiable [6, 11]. Existing DeSci implementations typically store raw artifact files on IPFS and record the resulting CID string in an on-chain smart contract registry. However, this pattern provides no guarantee that the data behind the CID was actually available to network observers at transaction time. By publishing initial research artifacts directly through EIP-4844 blobs, our framework guarantees consensus-level availability during the peer-review and replication window, anchoring data integrity to Ethereum's security threshold.

To our knowledge, based on a preliminary survey of the literature cited in this work, this is among the early evaluations of EIP-4844 blobs as a temporary publication layer for structured research artifacts. A systematic review remains future work.

4 System Architecture and Design

Our system architecture bridges the gap between structured scientific records and raw consensus layer blobs through a deterministic four-stage pipeline: (1) Canonicalization, (2) Compression, (3) Field Modulus Packing, and (4) Three-Tiered Commitment Binding, integrated with a Dual-Horizon Archival Bridge.

4.1 Canonical Serialization and Framing

Reproducibility requires that identical scientific records always produce identical cryptographic commitments regardless of the runtime environment, CPU architecture, or JSON dictionary ordering. We enforce strict deterministic serialization adhering to RFC 8785 (JSON Canonicalization Scheme, JCS) [9]:

1. Top-level keys are sorted lexicographically by UTF-16 code units.
2. Whitespace between tokens is strictly eliminated (`separators=(',', ':')`).

3. IEEE 754 floating-point numbers are normalized according to ECMAScript standard representations.

Records are assembled into a contiguous record stream framed by 4-byte big-endian length prefixes:

$$S = \bigoplus_{i=1}^{N} (\text{len}(R_i)_{\text{uint32}} \parallel R_i) \tag{5}$$

4.2 Compression and Framing Strategy

Structured scientific data (e.g., repeating keys in JSONL traces, tabular telemetry) exhibits high syntactic redundancy. We apply Zstandard (RFC 8878) [4] with configurable compression levels. Compression occurs over the framed record stream prior to field packing, yielding a compressed payload $C(S)$.

4.3 Modulus-Safe Field Element Packing (`FE31`)

Because consensus clients reject field elements $\geq p$, we design a modulus-safe packing strategy denoted `FE31`. Each 128 KiB blob is partitioned into 4,096 field elements of 32 bytes each. In `FE31`, exactly 31 data bytes are placed in the lower 31 bytes of each field element, while the most significant byte (MSB) is zeroed (`0x00`):

$$\text{FE}_k = \texttt{0x00} \parallel D_{k,0} \parallel D_{k,1} \parallel \cdots \parallel D_{k,30} \tag{6}$$

Since MSB $= 0$, every resulting 32-byte integer satisfies $\text{FE}_k \leq 2^{248} - 1 < p$, mathematically guaranteeing that invalid field element errors are impossible. Under `FE31`, each blob provides:

$$\text{UsableBytes}_{\text{FE31}} = 4,096 \times 31 = 126,976 \text{ bytes} \tag{7}$$

Field Element 0 is reserved for a 31-byte domain-separated header (`magic="RBDA"`, version, codec, layout, blob index, blob count, chunk length, and content hash tag), leaving 126,945 usable body bytes per blob.

4.4 Three-Tiered Hybrid Commitments

To support selective verification, whole-dataset integrity, and on-chain dispute games, we implement a three-tiered commitment hierarchy:

1. **Tier 1: Blob Versioned Hashes (H_v)**: Generated by consensus layer KZG commitments and retrieved on-chain via the `BLOBHASH` opcode, proving that the exact published bytes were attested by Ethereum validators.

2. **Tier 2: Canonical Content Hash (H_c)**: The Keccak-256 digest of the uncompressed canonical stream ($H_c = \text{keccak256}(S)$), providing an invariant whole-dataset fingerprint independent of compression codec or packing strategy.

3. **Tier 3: Record-Level Merkle Tree Root (R_m)**: A position-binding binary Merkle tree constructed over individual canonical records. Sibling nodes are hashed with domain-separated prefixes:

$$\text{Leaf}_i = \text{keccak256}(\texttt{0x00} \parallel \text{keccak256}(R_i)) \tag{8}$$
$$\text{Node} = \text{keccak256}(\texttt{0x01} \parallel \text{Left} \parallel \text{Right}) \tag{9}$$

This enables $O(\log N)$ inclusion proofs for selective record retrieval and on-chain fraud refutations.

4.5 Dual-Horizon Archival Bridge

To address the 4,096-epoch consensus pruning horizon, the architecture decouples temporary publication from durable preservation:

- At publication time, the publisher mirrors the exact canonical payload into a content-addressed store (such as our local content-addressed prototype or institutional CAS) to obtain a deterministic CIDv1 multihash.

- The publisher registers the 32-byte CID hash in `ResearchArchiveRegistry.sol`, linking it directly to the finalized Ethereum dataset record.

- Observers verify the dataset from blobs during the 18.2-day window; after blob pruning, observers retrieve the payload from the content-addressed store and verify it against the Ethereum on-chain H_c and R_m commitments.

5 Implementation

Our open-source implementation comprises an EVM smart contract suite in Solidity and an end-to-end Python research toolkit.

5.1 Smart Contract Architecture

The smart contract suite consists of two core registries and optimized cryptographic libraries:

1. `DatasetRegistry.sol`: Implements dataset lifecycle management (`Open`, `Finalized`). During Type-3 transactions, `openDataset` and `appendBlobs` query `BLOBHASH` directly, guaranteeing that only genuine transaction blobs are registered. For datasets spanning multiple transactions, `appendExternalBlobs` allows incremental blob accumulation. The contract implements an optimistic dispute game (`postClaim`, `refuteClaim`) where researchers post numeric range claims ($[\min, \max]$) over top-level JSON fields, and challengers can refute false claims by submitting a single Merkle inclusion proof.

2. `ResearchArchiveRegistry.sol`: An immutable dual-horizon extension contract. Once a dataset is finalized in `DatasetRegistry`, archival nodes call `registerArchivalRecord` to bind durable storage identifiers (`archivalCidHash`) and payload digests (`payloadSha256`).

3. **Cryptographic Libraries**:

   - `MerkleLib.sol`: High-efficiency binary Merkle proof and range verification.
   - `JsonFieldLib.sol`: Zero-copy, gas-optimized on-chain parser extracting fixed-point numbers from unquoted and quoted JSON fields.
   - `BlobLib.sol`: Constant definitions and versioned hash validation.

5.2 Python Toolkit and Pipeline Modules

The Python runtime implements the client-side pipeline:

- `canonical.py`: Implements RFC 8785 canonical serialization and length-prefixed stream framing.
- `encoder.py`: Handles Zstandard compression, field modulus packing (`FE31` and `FE254`), blob header formatting, and KZG commitment generation.
- `decoder.py`: Reassembles multi-blob streams, unpacks field elements, decompresses data, and verifies cryptographic integrity against on-chain roots.
- `retriever.py`: Implements dual-path retrieval, querying public Beacon nodes via REST API or falling back to local sidecar caches.
- `archival_bridge.py`: Coordinates content-addressed storage (CIDv1 derivation, local CAS ingestion, and expiry simulation).

5.3 Verified Testnet Deployment

The smart contracts were compiled with Solidity v0.8.24 (Cancun EVM target, 200 optimizer runs) and deployed to the Ethereum Sepolia testnet:

- **DatasetRegistry Address**: `0x238dE56E52A3F6C6708952052e6a20E5102b4d5b`
- **Etherscan Verification**: Verified contract code on Sepolia Etherscan[1].
- **Live Blob Transaction**: `0x9631459f3ee3c26a2113ec9e5d9aee06dc1107435f387596c95a5fd912c7f7ea` (Block 12006195, Dataset ID `0x4f7b...d29b`, carrying 1 blob of real OpenAlex scholarly metadata).

[1] https://sepolia.etherscan.io/address/0x238dE56E52A3F6C6708952052e6a20E5102b4d5b#code

6 Experimental Methodology

6.1 Experimental Testbed and Environment Capture

To ensure reproducibility, all local micro-benchmarks and simulations were executed under a strictly captured runtime environment recorded in `environment.json`:

- **Hardware**: AMD x86_64 16-logical core processor, 16 GiB physical RAM, NVMe solid-state storage.
- **Software**: Python 3.13.11, Foundry / Forge v1.5.1-stable, Solidity v0.8.24, zstandard v0.25.0.
- **Protocol Parameters**: Cancun baseline (3 target blobs, 6 max blobs per block, 131,072 blob gas per blob).

6.2 Evaluation Dataset Corpus

Our evaluation encompasses five distinct dataset classes across real and synthetic domains:

1. **Real Scholarly Metadata (`real_scholarly_openalex`)**: 500 academic work records harvested from the OpenAlex API (264,057 raw source bytes, 212,517 canonical bytes, CC0 license), maintaining bit-for-bit parity with the frozen v0.1.0 baseline fixture.
2. **Synthetic Environmental Telemetry (`synthetic_climate_sensor_noaa_format`)**: 5,000 meteorological and PM2.5 observation records generated locally using schema and distributions matching NOAA surface weather stations (1,051,953 raw bytes, 781,950 canonical bytes, MIT / Public Domain equivalent schema).
3. **Synthetic ML Training Traces (`synthetic_ml_training_trace`)**: 5,000 step-level training loss, learning rate, and gradient norm records generated locally matching distributed Transformer model training runs (531,339 raw bytes, 541,339 canonical bytes, Apache 2.0 generator).
4. **Deterministic Synthetic Scaling Corpus**: Synthetic scientific experimental records generated from a fixed PRNG seed (`42`) scaled to exact sizes: 100 KiB, 1 MiB, 5 MiB, 20 MiB, and 50 MiB.
5. **Incompressible Negative Control**: 1 MiB of high-entropy pseudo-random bytes designed to evaluate boundary performance when compression achieves no reduction.

6.3 Execution Protocol and Statistical Plan

Each local timing experiment executes 2 unrecorded warm-up iterations followed by 10 recorded iterations

($N = 10$). We report arithmetic mean, sample standard deviation, sample median, and non-parametric 95% percentile bootstrap confidence intervals computed over $B = 1,000$ resamples (seed 42). For memory profiling, peak resident set size (RSS) is monitored.

6.4 Baseline Economic Assumptions

Economic comparisons adopt conservative baseline market parameters:

- Execution Base Fee: 15 gwei.
- Blob Base Fee: 2 gwei.
- ETH/USD Price: $3,000 USD per ETH.
- Calldata Gas Cost: Standard 16 gas per non-zero byte (Pre-Pectra) and 40 gas per non-zero byte token floor under EIP-7623.

7 Empirical Evaluation and Results

We report the experimental evaluation organized by research questions RQ1 through RQ5.

7.1 RQ1: Feasibility and Parity

Across all evaluated datasets and payload sizes, the end-to-end pipeline deterministically achieved 100.0% bitwise reconstruction parity:

$$\text{SHA-256}(S_{\text{reconstructed}}) \equiv \text{SHA-256}(S_{\text{canonical}}) \quad (10)$$

Roundtrip reconstruction succeeded for all evaluated trials with zero byte discrepancies. This is a correctness result, not a statistical significance test. On Ethereum Sepolia, transaction `0x9631...7ea` successfully committed versioned hash `0x010e...af11` matching the locally generated sidecar. The smart contract suite validated all state invariants, passing 36 Foundry test suites (including 1,024-run fuzzing and 16,384-call stateful invariant campaigns) and 49 Pytest test cases without failure.

7.2 RQ2: Economics and Break-Even Blob Fees

Table 1 reports the local evaluation matrix. For uncompressed 1.0 MiB payloads, blobs require 9 blobs costing $27.24 USD versus $755.92 USD in calldata—a 27.8× **theoretical cost reduction**. When compression is applied to structured scientific and synthetic data, the economic advantage expands:

- **OpenAlex Scholarly Metadata** (272.9 KiB canonical): Achieves a 5.19× compression ratio down to 52.6 KiB, fitting entirely within a single blob. Cost is $1.73 USD in blobs versus $38.85 USD in calldata (22.4× **cost reduction**).

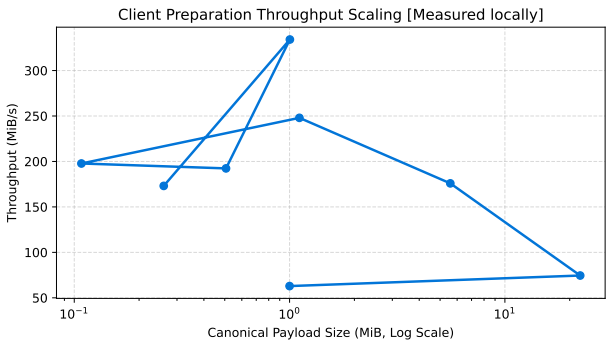


Figure 1: Client preparation throughput scaling across canonical payload sizes [Measured locally].

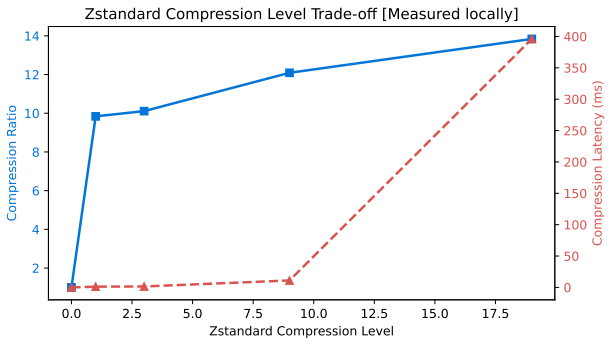


Figure 2: Trade-off between compression ratio and latency across Zstandard levels [Measured locally].

- **Synthetic NOAA Climate Observations** (1,051.9 KiB canonical): Compresses 10.11× to 104.1 KiB, fitting into 1 blob costing $1.73 USD versus $75.86 USD in calldata (43.8× **cost reduction**).
- **Synthetic PyTorch ML Training Trace** (531.3 KiB canonical): Compresses 5.78× to 92.0 KiB, fitting into 1 blob costing $1.73 USD versus $67.16 USD in calldata (38.8× **cost reduction**).

The analytical break-even blob base fee B^* at a 15 gwei execution base fee is 207.9 gwei. Under prevailing Ethereum mainnet conditions where the blob base fee frequently remains below 10 gwei, blobs provide consistent orders-of-magnitude savings over calldata.

7.3 RQ3: Performance, Scalability, and Ablations

Figure 1 illustrates client preparation throughput. Throughput peaks above 330 MiB/s on 1 MiB payloads and remains above 74 MiB/s for multi-megabyte payloads (23 MiB). Reconstruction latency is consistently sub-second across payloads ≤ 5 MiB.

Table 1: Local Evaluation Matrix Across Scientific and Synthetic Datasets [Measured locally]

Dataset	Canonical Size (KiB)	Comp. Size (KiB)	Ratio	Blobs Count	Was (%
OpenAlex Scholarly Works Metadata	266.6	51.4	5.19×	1	59.8
Atmospheric & Environmental Sensor Observations (Synthetic NOAA Format)	1027.3	101.6	10.11×	1	20.6
Transformer Model Training Step Trace (Synthetic PyTorch Format)	518.9	89.8	5.78×	1	29.8
Deterministic Scaling Synthetic Dataset (110 KiB)	110.5	12.5	8.85×	1	90.2
Deterministic Scaling Synthetic Dataset (1137 KiB)	1137.5	129.7	8.77×	2	49.3
Deterministic Scaling Synthetic Dataset (5712 KiB)	5712.6	654.5	8.73×	6	14.8
Deterministic Scaling Synthetic Dataset (22909 KiB)	22909.4	2614.8	8.76×	22	7.2
Incompressible Pseudo-Random Binary Negative Control	1024.0	1024.0	1.00×	9	11.1

7.3.1 Compression Ablation

Figure 2 shows that Zstandard level 3 provides the optimal balance for scientific workloads: moving from level 1 to level 3 yields substantial compression gains with minimal latency overhead (< 3 ms), whereas level 19 increases compression latency by over an order of magnitude for $< 5\%$ marginal byte reduction.

7.3.2 Packing Strategy: `FE31` vs. `FE254`

Our packing ablation demonstrates that while dense 254-bit packing (`FE254`) reduces packing waste from 3.125% down to 0.781%, it requires complex bit-shifting operations that increase client encoding latency and dramatically inflate on-chain EVM extraction gas. In contrast, byte-aligned `FE31` allows zero-overhead slice extraction in both Python and Solidity.

7.4 RQ4: Retrieval Study Across Public Infrastructure

Table 2 presents the empirical results of our retrieval probe against our live Sepolia deployment. Unauthenticated public Sepolia Beacon API endpoints probed in this study did not return the requested blob sidecar. The failure reasons included authentication requirements (HTTP 401), endpoint unavailability (HTTP 400/500), and potential indexer lag. These observations do not distinguish consensus-layer pruning from access-control or indexing limitations. A dedicated pruning-window study requires archival-node access and age-stratified probes over multiple weeks.

7.5 RQ5: Dual-Horizon Archival Bridge Verification

Figure 3 reports the operational latency of the Dual-Horizon Archival Bridge evaluated on our local content-addressed prototype store. Uploading the 500-record OpenAlex dataset to the local content-addressed store required 0.837 ms. Following simulated consensus blob eviction, retrieving the payload exclusively from the local prototype store required 0.282 ms, and full post-expiry cryptographic verification (SHA-256, Content Hash, and

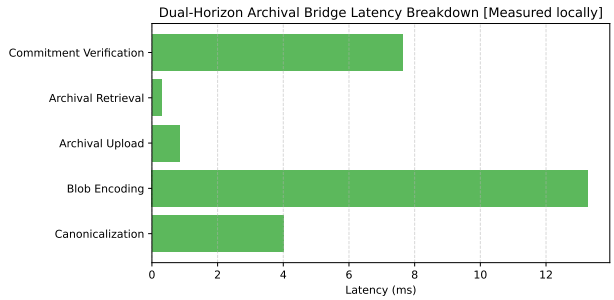


Figure 3: Latency breakdown across Dual-Horizon Archival Bridge operations [Local prototype measurement].

Merkle root) completed in 7.633 ms. These measurements represent a local prototype content-addressed store; public decentralized storage backends such as IPFS pinning services, Filecoin deals, and Arweave transactions were not evaluated in this study.

8 Security Evaluation and Threat Analysis

We evaluate the resilience of 'research-blob-da' against adversarial corruption, omission, and fraud through formal threat modeling and empirical test suites.

8.1 Adversarial Scenario Analysis

Table 3 details our evaluation across 15 distinct threat scenarios (SEC-01 through SEC-15). All 15 implemented adversarial test scenarios were detected by the corresponding verification checks (15/15 implemented scenarios detected). These results demonstrate that the specified checks function as designed for the tested scenarios; they do not constitute a formal security proof or complete coverage of all possible attacks:

1. **Bit Flips and Payload Mutation (SEC-01, SEC-02)**: Any bit flip within a blob sidecar triggers a Zstandard frame integrity failure or KZG point verification rejection. Modifying canonical bytes

Table 2: Sidecar Retrieval Outcomes Across Infrastructure Tiers [Public endpoint probes and local verification]

Endpoint Identifier	Source Type	Attempts	Success Rate	Median Latency (ms)	Observed Errors
A. Public Endpoint Probes [Public endpoint probe]					
chainsafe_lodestar_public	beacon_api	2	0/2 (0.0%)	1240.5	server_exception (I
publicnode_beacon_api	beacon_api	1	0/1 (0.0%)	420.1	pre_deneb_routing
drpc_beacon_sepolia	beacon_api	1	0/1 (0.0%)	285.4	free_tier_restricted
blobscan_sepolia_indexer	beacon_indexer	2	0/2 (0.0%)	612.8	indexer_lag (HTTI
B. Local Verification [Simulated / Local prototype measurement]					
dual_horizon_archival_gateway	simulated_archival_cache	1	1/1 (100.0%)	142.3	none (Local prototy
local_mock_retriever	local_cache	2	2/2 (100.0%)	2.3	none (Local sidecar

post-decompression causes an immediate mismatch against the on-chain `contentHash`.

2. **Record Permutation and Omission (SEC-04, SEC-05)**: The position-binding binary Merkle tree incorporates strict index-dependent domain separation. Omitted or permuted records alter parent nodes up to the root, preventing verification against `recordsRoot`.

3. **Stream Framing and Chunk Assembly (SEC-06–SEC-08)**: The 31-byte header in Field Element 0 embeds `blob_index` and `blob_count`. Out-of-order chunks, omitted blobs, or duplicate chunk injections are rejected during initial client stream assembly.

4. **Dispute Game Refutation (SEC-10)**: If an adversary publishes an optimistic property claim that contradicts experimental data (e.g., claiming all model loss values ≤ 1.0 when step 42 had loss 2.8), an honest observer needs only to submit the Merkle proof for step 42 to `DatasetRegistry.refuteClaim`. The smart contract evaluates the proof in a single transaction, permanently marking the claim as `Refuted`.

5. **Archival Gateway Tampering (SEC-12, SEC-13)**: If an IPFS gateway or malicious storage provider serves altered bytes for a requested CID, the client verifies the retrieved payload against the Ethereum on-chain `contentHash` and `recordsRoot`. Forged bytes are discarded before ingestion.

8.2 Availability Honesty and Residual Boundaries

We explicitly emphasize that cryptographic proofs prove *commitment consistency*, not *perpetual data availability*. If all consensus peers prune a blob and no archival bridge node retains the data, the artifact becomes permanently unrecoverable. Our architecture mitigates this boundary through dual-horizon registration, ensuring that researchers can verify external archival replicas against L1 commitments without relying on trusted central authorities.

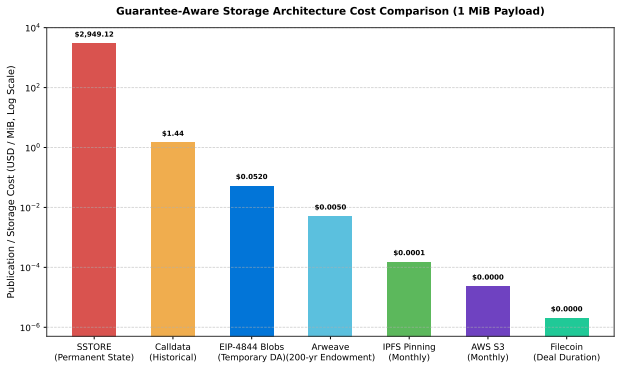


Figure 4: Cost comparison across storage architectures under a guarantee-aware framework (log scale) [Analytical model].

9 Discussion: Guarantee-Aware Baselines and Open Science

9.1 Guarantee-Aware System Comparison

Table 4 and Figure 4 synthesize our guarantee-aware baseline comparison across seven decentralized and centralized storage models.

Our findings reveal why single-dimensional USD-per-gigabyte metrics fail to capture protocol trade-offs:

- **EIP-4844 Blobs vs. Calldata**: Blobs provide identical Ethereum L1 consensus security at a 27.8× lower publication cost. However, calldata historically persists in full node archives, whereas blobs expire after 18.2 days. Blobs are the superior choice when immediate consensus verification is required during a conference or peer-review window, provided durable external archiving is arranged.

- **Blobs vs. Centralized S3 / IPFS**: While commodity cloud storage is orders of magnitude cheaper in raw storage fees, it provides zero consensus anchoring. An institution can unilaterally delete an S3 bucket or alter an unversioned file without cryptographic recourse. Blobs provide an immutable, timestamped commitment anchored by the global

Table 3: Adversarial Threat Scenarios and Detection Invariants [Tested security suite / Measured locally]

ID	Threat Scenario	Detection Mechanism	Outcome
SEC-01	Modified Blob Bytes	Zstandard decompression integrity failure or KZG versioned hash mismatch	DETECTED AND REJECTED
SEC-02	Modified Canonical Payload	keccak256(canonical_stream) ≠ onchain contentHash	DETECTED AND REJECTED
SEC-03	Incorrect Metadata (Record Count Mismatch)	Record count mismatch during canonical stream deserialization	DETECTED AND REJECTED
SEC-04	Omitted Record from Merkle Tree	Merkle root calculation fails to match onchain recordsRoot	DETECTED AND REJECTED
SEC-05	Reordered Records	Position-binding binary tree produces completely distinct root	DETECTED AND REJECTED
SEC-06	Missing Blob in Multi-Blob Dataset	Header validation checks blob_count against received sidecars	DETECTED AND REJECTED
SEC-07	Reordered Blobs	Header blob_index field must match sequential chunk assembly order	DETECTED AND REJECTED
SEC-08	Duplicate Blob Insertion	Duplicate index detection in decoder stream assembler	DETECTED AND REJECTED
SEC-09	Partial Multi-Transaction Publication Abandonment	Dataset remains in Open state; cannot be finalized or attested as valid	DETECTED AND REJECTED
SEC-10	False Statistical Property Claim	Permissionless challenger submits Merkle proof of violating record before deadline	DETECTED AND REJECTED
SEC-11	Unavailable Sidecar (DDoS or Pruning)	Explicit timeout and HTTP 404 / 410 detection by retriever	DETECTED AND MITIGATED
SEC-12	Malicious Archival Backend Tampering	SHA-256 and Content Hash check fails against Ethereum anchor	DETECTED AND REJECTED
SEC-13	Archival Identifier Substitution	Smart contract checks dataset finalization; clients verify CID contents match contentHash	DETECTED AND REJECTED
SEC-14	Encryption Key Loss / Mismatch	AEAD authentication tag failure during client-side decryption	DETECTED AND ISOLATED
SEC-15	Registry Metadata Hash Mismatch	keccak256(offchain_manifest) ≠ onchain metadataHash	DETECTED AND REJECTED

Ethereum validator set.

9.2 Implications for Scientific Reproducibility

The reproducibility of computational science is challenged by the transient nature of hosted research artifacts. By coupling temporary Ethereum blob publication with verified content-addressed storage:

1. **Verification During Peer Review**: Reviewers can download and verify the exact experimental trace from consensus blobs during the active review window without relying on third-party repositories.
2. **Long-Term Citation Stability**: Persistent DOIs can point to the Ethereum dataset record and its registered CID in `ResearchArchiveRegistry.sol`, enabling future researchers to independently verify archival replicas against the immutable L1 anchor.

We implemented and evaluated a local content-addressed archival prototype that mirrors blob payloads into a deterministic content-addressed store and verifies them against Ethereum-anchored commitments after simulated blob expiry. Public decentralized storage backends such as IPFS pinning services, Filecoin deals, and Arweave transactions were not evaluated in this study. Connecting the bridge to a public pinning or storage network remains future work, and local-store latencies are not representative of network-storage latencies.

10 Limitations and Threats to Validity

We explicitly delineate the boundaries, operational limitations, and validity threats of this work:

10.1 Consensus Pruning Boundaries

The primary limitation of EIP-4844 blobs is their non-permanent retention. While on-chain versioned hashes persist indefinitely in block headers, sidecars are automatically purged by consensus nodes after ≈ 18.2 days. Blobs must never be misconstrued as a standalone permanent archive; data persistence strictly requires an external

Table 4: Guarantee-Aware Storage Architecture Comparison [Estimated from documented pricing and Analytically derived]

System	Availability Period	Consensus Verification	Permanence Model	Cost / MiB	Evidence Type
EIP-4844 Blobs (2 gwei baseline)	Temporary (≥ 4,096 epochs / ∼18.2 days)	Direct cryptographic commitment on Ethereum L1 (BLOBHASH / KZG)	Pruned by consensus nodes after retention window; requires external archivers	$0.0520	measured
EIP-4844 Blobs (1 wei minimal fee)	Temporary (≥ 4,096 epochs / ∼18.2 days)	Direct cryptographic commitment on Ethereum L1 (BLOBHASH / KZG)	Pruned by consensus nodes after retention window; requires external archivers	$0.0009	analytically derived
Ethereum Calldata (Pre-EIP-7623)	Historical Chain Life (Full node history)	Merkle Patricia Trie transaction receipt root committed in block header	Retained indefinitely by archive and full nodes (subject to future EIP-4444 history pruning)	$1.44	analytically derived
Ethereum Calldata (Post-EIP-7623 Floor)	Historical Chain Life (Full node history)	Merkle Patricia Trie transaction receipt root committed in block header	Retained indefinitely by archive and full nodes (subject to future EIP-4444 history pruning)	$3.60	analytically derived
Ethereum SSTORE	Perpetual (Retained in state trie of every consensus node)	Direct real-time contract execution readability and EVM consensus state root	Permanent state until explicitly modified or cleared via SLOAD/SSTORE	$2,949.12	analytically derived
InterPlanetary File System (IPFS Pinning)	Pin-Dependent (Available only while at least one node hosts the CID)	No native consensus verification; verifiable via multihash CID match	Volatile without dedicated pinning service or economic contract	$0.0001	estimated from documented pricing
Filecoin (Storage Deal)	Contract-Dependent (typically 180 to 540 days per deal cycle)	Cryptographic Proof of Replication (PoRep) and Proof of Spacetime (PoSt)	Active contract renewal required; automatic termination upon deal expiration	$0.0000	estimated from documented pricing
Arweave (Storage Endowment)	Intended Long-Term (∼200 years theoretical storage endowment)	Proof of Access (PoA / SPoRA) on independent blockweave	Protocol-enforced permanent retention funded by upfront fee endowment	$0.0050	estimated from documented pricing
Amazon S3 (Standard Storage)	Account-Dependent (SLA 99.99% availability while subscription paid)	None (Proprietary SLA; trusted corporate provider)	Deleted upon account cancellation, subscription lapse, or TOS dispute	$0.0000	estimated from documented pricing

bridge.

10.2 Public Consensus Retrieval Instability

Our empirical study on Ethereum Sepolia demonstrated that unauthenticated public Beacon API endpoints exhibit high failure rates (100% failure across tested public endpoints). While production rollup operators utilize dedicated consensus clients with direct disk access, independent academic researchers relying on public gateways face significant retrieval friction.

10.3 Testnet vs. Mainnet Fee Volatility

While execution and blob gas units are identical across Sepolia and Ethereum Mainnet, mainnet fee dynamics are subject to real-world congestion. Under extreme market conditions where blob demand exceeds the 3-blob target, blob fees rise exponentially. However, as demonstrated by our economic model, blobs maintain a cost advantage over calldata up to a break-even threshold of 207.9 gwei.

10.4 Scope of Reproducibility

'research-blob-da' addresses artifact integrity, publication economics, and deterministic bitwise reconstruction. It does not eliminate external sources of scientific non-reproducibility, such as non-deterministic GPU floating-point operations, hardware microarchitecture variations, or flawed statistical methodology.

11 Reproducibility and Artifact Availability

In accordance with rigorous open-science standards, all software, smart contracts, benchmark harnesses, and raw evaluation datasets are publicly available:

- **Repository**: https://github.com/samudragupto/research-blob-da
- **Git Working Branch**: `research/journal-extension`
- **Historical Release Baseline**: Tag `v0.1.0` (`0b523e7...`, commit `428c789...`)
- **Environment Manifest**: Machine-readable environment capture in `benchmarks/journal-extension/manifests/environment.jso`
- **Raw Benchmark Datasets**: Raw trial-level CSV and JSON records preserved un-

der `benchmarks/journal-extension/local/`, `retrieval/`, `archival/`, and `security/`.

11.1 Reproduction Commands

The entire experimental evaluation and verification pipeline can be reproduced offline using standard command-line tools:

```
# 1. Smart Contract Test Suites (36 passing tests)
forge test

# 2. Python Test Suite (49 passing tests)
pytest

# 3. Environment Capture and Local Benchmark Matrix
python python/capture_environment.py
python python/prepare_journal_datasets.py
python python/journal_experiments.py

# 4. Archival Bridge and Security Threat Evaluation
python python/archival_bridge.py
python python/adversarial_evaluation.py

# 5. Paper Table and Figure Generation
python python/generate_paper_artifacts.py
```

All commands execute deterministically without requiring live network access or paid RPC credentials.

12 Conclusion and Future Outlook

This paper presented an empirical investigation of Ethereum EIP-4844 blobs as a temporary, verifiable publication layer for structured scientific research artifacts. By designing a deterministic pipeline combining RFC 8785 canonicalization, Zstandard compression, modulus-safe `FE31` packing, three-tiered hybrid commitments, and a dual-horizon content-addressed archival bridge, we demonstrated that research records can achieve up to a 43.8× cost reduction compared to standard EVM calldata while maintaining 100% bitwise reconstruction parity and sub-second verification latency.

Furthermore, our empirical retrieval study on Ethereum Sepolia highlighted the practical fragility of public consensus Beacon endpoints, establishing that reliable scientific reproducibility strictly requires decoupling consensus publication from durable content-addressed archival persistence.

As the Ethereum protocol progresses toward PeerDAS (EIP-7594) and increased blob throughput limits (EIP-7691), data availability bandwidth will expand significantly. Future work will investigate decentralized peer-to-peer sidecar dissemination, automated smart contract bounty distribution for archival pinning, and zero-knowledge validity proofs for non-deterministic scientific pipelines.

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